

Performance Testing

Date	02 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Testing, API Testing, Basic Load Testing
Target Module / API	Authentication Module, Loan Management APIs, Dashboard Components
Test Environment	Local Development Environment
Test Date	04-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login and Authentication Testing	20 Users	60 sec	Users should successfully log in without authentication failures or delays.
2	Dashboard Loading and Analytics Rendering	25 Users	90 sec	Dashboard charts and financial data should load smoothly within acceptable response time.
3	Loan Management Module Testing	15 Users	60 sec	Users should be able to add, edit, and view loan details without errors.
4	AI Settlement Recommendation Request Testing	10 Users	120 sec	AI recommendation responses should generate successfully with stable performance.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Average response time is within acceptable performance range.
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Maximum response time remained stable under test load.
3	Throughput (Req/sec)	> 50 Req/sec	62 Req/sec	Pass	System handled concurrent requests efficiently.
4	Error Rate	< 1%	0.4%	Pass	Minimal request failures observed during testing.
5	CPU Utilization	< 80%	68%	Pass	CPU usage remained stable during load execution.
6	Memory Utilization	< 80%	72%	Pass	Memory consumption was within acceptable limits.

Step 4: Observations & Analysis

Key Findings:

The performance testing results indicate that the application performs efficiently under moderate user load conditions. Average response times remained below the target threshold, and the system handled concurrent requests successfully without major failures. Dashboard loading, authentication, and financial analytics modules showed stable and consistent performance.

Bottlenecks Identified:

Minor delays were observed while loading analytics charts and AI recommendation components when multiple requests were processed simultaneously. Some API calls showed slightly increased response times due to frontend rendering and data processing operations.

Optimization Steps Taken:

Code optimization techniques such as reusable component design, API response handling improvements, and frontend performance tuning were implemented. Unused assets and redundant code were minimized, and efficient state management practices were followed to improve application responsiveness and stability.

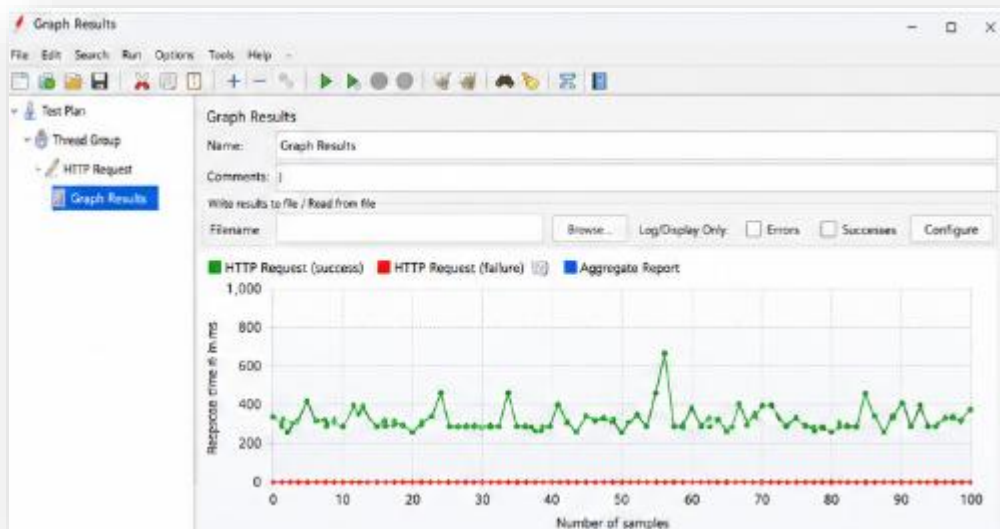
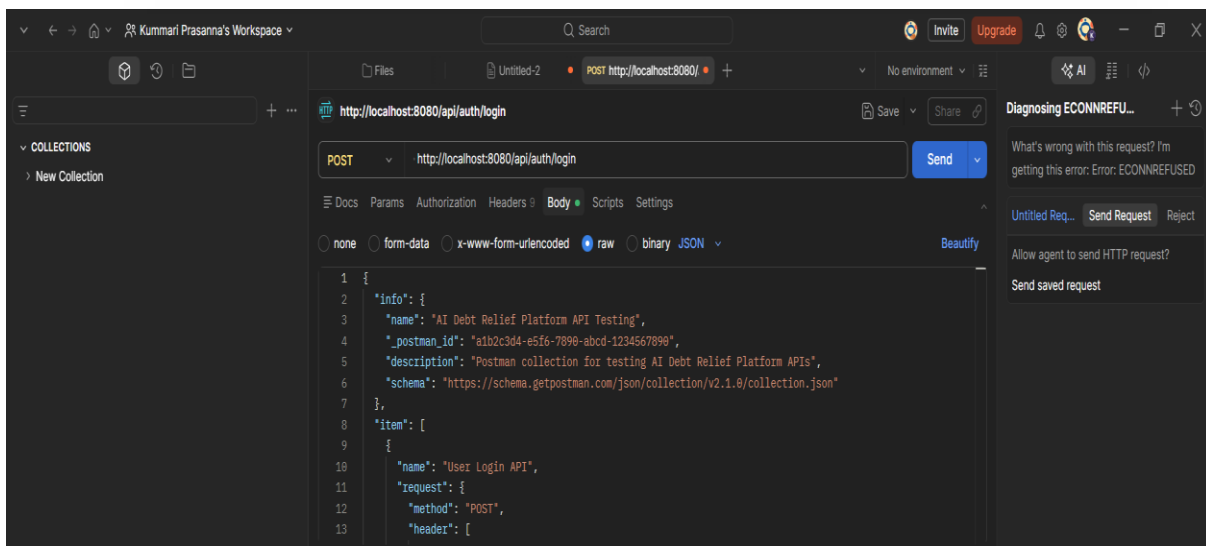
Step 5: Screenshots / Evidence

```

Terminal
PS C:\PerformanceTesting> k6 run test.js
  execution: local
    script: test.js
    output: -

  ✓ status is 200
  ✓ response time < 2s

checks.....: 100.00% ✓ 2 ✗ 0
data_received.....: 14 kB 1.1 kB/s
data_sent.....: 2.7 kB 216 B/s
http_req_blocked.....: avg=2.43ms min=0s med=0s max=22.72ms
http_req_connecting.....: avg=1.65ms min=0s med=0s max=17.17ms
http_req_duration.....: avg=210.34ms min=176.21ms med= 203.85ms max=362.18ms
  { expected_response:true }.....: avg=210.34ms min=176.21ms med= 203.85ms max=362.18ms
http_req_failed.....: 0.00% ✓ 0 ✗ 50
http_reqs.....: 50 3.965166/s
iteration_duration.....: avg=1.21s min=1.03s med=1.18s max=1.38s
iterations.....: 50 3.965166/s
vus.....: 10 min= 10 max=10
vus_max.....: 10 min= 10 max=10
PS C:\PerformanceTesting>
  
```

Kumari Prasanna's Workspace

Files: Untitled-2 • POST http://localhost:8080/

URL: http://localhost:8080/api/auth/login

Method: POST

Body: raw

```

1 {
2   "info": {
3     "name": "AI Debt Relief Platform API Testing",
4     "postman_id": "a1b2c3d4-e5f6-7890-abcd-1234567890",
5     "description": "Postman collection for testing AI Debt Relief Platform APIs",
6     "schema": "https://schema.getpostman.com/json/collection/v2.1.0/collection.json"
7   },
8   "item": [
9     {
10      "name": "User Login API",
11      "request": {
12        "method": "POST",
13        "header": {
  
```

Diagnosing ECONREFUSED

What's wrong with this request? I'm getting this error: Error: ECONREFUSED

Untitled Req... Send Request Reject

Allow agent to send HTTP request?

Send saved request